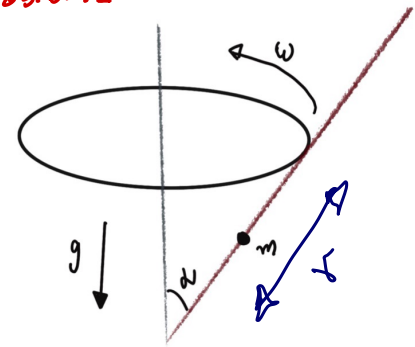
1 A bead of mass m slides on a long straight wire which makes an angle α with, and rotates with constant angular velocity ω about, the upward vertical. As shown in the figure below. Gravity g acts vertically downwards!

Problem 1



1. Choose an appropriate genaralised coordinate and finad the Lagrangian.
2. Write down the Lagrange's equation of motion.
3. Write down the effective potential and find the stable equilibrium points if any

motion is only along wire and we can choose r along that both $\dot{\theta}$ and $\dot{\phi}$ are zero, but there is rotation

$$T = \frac{m}{2}\left[\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right]$$

first set $\dot{\theta} = 0$ $\quad \dot{\phi} = \dot{\phi} + \omega$ ($\dot{\phi} = 0$)

$\theta = \alpha$

you can set $\dot{\theta} = \dot{\phi} = 0$ in rotating coordinate

$$T = \frac{m}{2}\left[\dot{r}^2 + r^2\sin^2\alpha\,\omega^2\right] \qquad V = mgr\cos\alpha$$

$$\mathcal{L} = \frac{m}{2}\left[\dot{r}^2 + r^2\sin^2\alpha\,\omega^2\right] - mgr\cos\alpha$$

$$EOM = m\ddot{r} = mr\sin^2\alpha\,\omega^2 - mg\cos\alpha$$

$$E = \frac{m}{2}\left[\dot{r}^2 + r^2 \sin^2\alpha \cdot \omega^2\right] + mgr\cos\alpha$$

$$0 = \cancel{m}\left[\cancel{\dot{r}}\ddot{r} + r\cancel{\dot{r}}\sin^2\alpha\,\omega^2\right] + \cancel{m}g\cancel{\dot{r}}\cos\alpha$$

$$\ddot{r} + r\sin^2\alpha\,\omega^2 + g\cos\alpha = 0$$

E defined as above is not conserved energy, we have to explicitly compute energy function

$$\dot{r}\frac{\partial L}{\partial \dot{r}} - L = h$$

$$m\dot{r}^2 - \frac{m}{2}\dot{r}^2 - r^2 m\sin^2\alpha\,\omega^2 + mgr\cos\alpha$$

$$h = \frac{m}{2}\dot{r}^2 - r^2 m\sin^2\alpha\,\omega^2 + mgr\cos\alpha$$

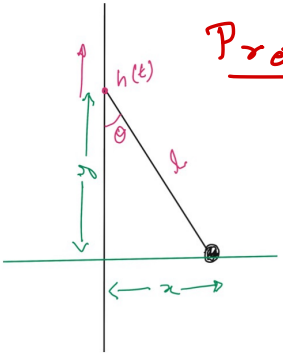
EOM is

$$\cancel{m}\ddot{r}\cancel{\dot{r}} - \cancel{2}r\cancel{\dot{r}}\,m\sin^2\alpha\,\omega^2 + mg\cancel{\dot{r}}\cos\alpha = 0$$

$$\ddot{r} - mr\sin^2\alpha\,\omega^2 + mg\cos\alpha = 0$$

Same as LEOM !

The point of support of a simple plane pendulum moves vertically according to $y = h(t)$, where $h(t)$, is an arbitary given function t. Gravity g acts vertically downwards along y.



Problems 2

1. Find the Lagrangian, taking as generalized coordinate the angle θ the pendulum.
2. Write down Euler-Lagrange's equation of motion.
3. Can gravity, g be compensated by $h(t)$

$$x = l \sin\theta \qquad y = h(t) - l\cos\theta$$

$$\dot{x} = l\cos\theta\,\dot{\theta} \qquad \dot{y} = \dot{h} + l\sin\theta\,\dot{\theta}$$

$$T = \frac{1}{2} m \left(l\dot{\theta}^2 + 2\dot{h} l \dot{\theta} \sin\theta + \dot{h}^2 \right)$$

$$V = mg(h - l\cos\theta)$$

$$\mathcal{L} = \frac{1}{2} m \left(l\dot{\theta}^2 + 2\dot{h} l \dot{\theta} \sin\theta + \dot{h}^2 \right) - mgh + mgl\cos\theta$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m(l\dot{\theta} + \dot{h} l \sin\theta)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = m(l\ddot{\theta} + \ddot{h} l \sin\theta + \dot{h} l \cos\theta\,\dot{\theta})$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = m\dot{h} l \dot{\theta} \cos\theta - mgl\sin\theta$$

$$m(l\ddot{\theta} + \ddot{h} l \sin\theta + \cancel{\dot{h} l \cos\theta\,\dot{\theta}}) = \cancel{m\dot{h} l \dot{\theta} \cos\theta} - mgl\sin\theta$$

$$m l^2 \ddot{\theta} = -m(g+\ddot{h})\, l \sin\theta$$

If $\ddot{h} = -g$ we can get rid of g!

Prob 3

For the potential given by the function,

$$V = -\frac{e^{-\alpha r}}{r},$$

find the effective radial potential and equilibrium points.

$$\mathcal{L} = \frac{m}{2}\left[\dot{r}^2 + r^2\dot{\phi}^2\right] + \frac{e^{-\alpha r}}{r}$$

$$m r^2 \dot{\phi} = L_z \longrightarrow L_z \text{ is constant}$$

$$\dot{\phi} = \frac{L_z}{m r^2}$$

$$E = \frac{m}{2}\left[\dot{r}^2 + \frac{L_z^2}{m^2 r^2}\right] - \frac{e^{-\alpha r}}{r}$$

$$\frac{m}{2}\left(E + \frac{e^{-\alpha r}}{r} - \frac{2L_z^2}{m r^2}\right) = \dot{r}^2$$

$$V = \frac{2L_z^2}{m r^2} - \frac{e^{-\alpha r}}{r}$$

$$\frac{\partial V}{\partial r} = -\frac{4L_z^2}{m r^3} + \frac{e^{-\alpha r}}{r^2} + \frac{\alpha e^{-\alpha r}}{r} = 0$$

$$\frac{\partial V}{\partial r} = -\frac{4L_z^2}{mr^3} + \frac{e^{-\alpha r}}{r^2} + \frac{\alpha e^{-\alpha r}}{r} = 0$$

$$\Rightarrow 0 = -\frac{4L_z^2}{m} + r e^{-\alpha r} + \alpha r^2 e^{-\alpha r} = 0$$

$$0 = -\frac{4L_z^2}{m} e^{\alpha r} + r + \alpha r^2$$

Let $e^{\alpha r} \approx 1$

$$0 = -\frac{4L_z^2}{m} + r + \alpha r^2$$

now where are roots

$$-1 \pm \sqrt{1 + 16\frac{L_z^2}{m}\alpha}$$

or something like this ?